Saransh Yadav

ma24btech11022@iith.ac.in | yadavsaransh06@gmail.com | +91 7817880691 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.31
Defence Public School, Dehradun, Uttarakhand	Class XII (CBSE)	2024	94.00%
DSB International Public School, Rishikesh, Uttarakhand	Class X (CBSE)	2022	98.17%

SCHOLASTIC ACHIEVEMENTS

- Campus Finalist in **Optiver Integral Cup**
- **AIR-4** in LIMITS (Category C: UG+PG), a national mathematics competition held by **ISI Bangalore** and **Jane Street**
- Consistently placed under **100** internationally in **SOF's International Mathematics Olympiad**

PROJECTS

Searching over Metric Spaces (DSA Lab Coursework) | Github Link

- Implemented and benchmarked **metric-space nearest neighbor search data structures**, including Vantage Point Trees, Generalized Hyperplane Trees (GHT), GNAT, M-Tree, Slim Tree, and Burkhard-Keller Trees among others.
- Evaluated search performance across datasets with varying size and dimensionality, comparing **partitioning** and **distance matrix** based indexing techniques.
- Analyzed indexing cost, query time, and distance computations to identify performance trade-offs; observed **GHT** excelled on high-dimensional datasets, **Excluded Middle VP Trees** as the fastest ball-partitioning approach, and **Fixed Query Array** exhibiting improved performance with increasing dimensionality.

Custom SQL-like Query Engine for Non Euclidean Metric Spaces | Github Link

- Built a query engine supporting standard SQL queries in addition with **custom similarity queries** to return top k results based on **semantic similarity**. Works for any dataset (need not be a vector space) with a meaningful metric.
- Implemented Context Free Grammar (**CFG**) and Abstract Syntax Tree (**AST**) for the custom similarity query
- Used **Generalized Hyperplane Trees** for indexing and integrated it using PyBind11 for good performance in higher dimensional datasets
- Benchmarked word embedding-based retrieval against direct metric-space search on **DNA data**, observing that edit-distance based retrieval achieved **5× higher Recall**@k (100% vs 20%) while also running as fast as the embedding baseline on the evaluated dataset

Hybrid Precision Plotting with Optimized Rendering | Github Link

- Built a high performance function plotter using recursive **adaptive sampling** to **reduce redundant computations**
- Optimized end-to-end throughput using **multi-threaded parallel dispatch**, thread-safe memoization, and compiler-driven **SIMD auto-vectorization**
- Implemented a **hybrid precision pipeline** triggered when numerical variance exceeds a tolerance, maintaining **96%+** of **computations** at **native hardware speed** while ensuring mathematical rigor near singularities and asymptotes
- Achieved a **~50x throughput improvement** in sub-microsecond evaluation cost per point over traditional plotting methods, **falling from 7.50 to 0.14-0.31 microseconds per point**

Text Summarization using TextRank and Deep Learning Methods | Github Link

- Built an abstractive and extractive text summarization pipeline using **Seq2Seq models**, **attention mechanisms**, and graph-based ranking algorithms namely **TextRank** on the **Xsum dataset**
- Evaluated summarization quality using **ROUGE metrics**, where the custom Seq2Seq-based abstractive model achieved **ROUGE-1/ROUGE-L scores of 0.8824**, outperforming the TextRank-based extractive baseline

(**0.2400/0.1067**) in **summary coherence** and **semantic retention**

EXTRACURRICULAR

Reached the **Screening Round** in **ISRO's Bharatiya Antariksh Hackathon 2025 | Github Link**

- Built a model for **automated monitoring of Himalayan glacial lakes** using Sentinel-1/2, Landsat-8/9 satellite imagery.
- Implemented an **EfficientNet-B3 + DeepLabV3+ segmentation pipeline** for glacial lake extraction and integrated **ConvLSTM**-based forecasting to predict future lake evolution and assess GLOF risk, returning an IOU of 95.6%

Placed **2nd** in **E-Summit Hackathon 2025** in partnership with **National Payments Corporation of India | Github Link**

- Developed an economically viable system where customers can share reward points across various merchants
- Presented a fully working prototype application where customers can choose where to spend their reward points

SKILLS

- **Programming/Scripting Languages:** C/C++, Python, Dart, SQL, LaTeX
- **Libraries/Frameworks:** Flutter, Pandas, Numpy, Scikit-Learn

RELEVANT COURSES

- **Computer Science / Engineering :** Data Structures and Algorithms, DBMS-I, Compilers-I, Theory of Computation
- **Mathematics :** Probability Theory, Applied Statistics, Convex Optimization, Measure Theory, Numerical Analysis, Ordinary Differential Equations

POSITIONS OF RESPONSIBILITY

Teaching Assistant | MA1230 - Series of Functions

- Assisted in the instruction of an undergraduate real analysis course
- Prepared assignments and supported students in topics including pointwise and uniform convergence, Taylor series, and an introduction to Fourier series.

Assistant Manager: Public Relations and Networking, ECell IIT Hyderabad